Calculating HFC and PFC Emissions from the Manufacturing, Installation, Operation and Disposal of Refrigeration & Air-conditioning Equipment (Version 1.0)

Guide to calculation worksheets (January 2005)

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I. Overview

I.A. Purpose and Domain of this Section

This guideline is written for plant managers and site personnel to facilitate the calculation and reporting of direct hydrofluorocarbon (HFC) emissions resulting from manufacturing, servicing, and disposal of refrigeration and air-conditioning equipment. Direct HFC emissions occur from sources that are owned or controlled by the company. A step-by-step approach is presented to cover every phase of the calculation process from data gathering to reporting.

This sector guideline should be applied to projects whose operations involve the manufacture, use, and disposal of refrigeration and air-conditioning equipment.

I.B. Process Description and Assumptions

Refrigeration and air-conditioning is composed of many end-uses, including household refrigeration, domestic air conditioning and heat pumps, mobile air conditioning, chillers, retail food refrigeration, cold storage warehouses, refrigerated transport, industrial process refrigeration, and commercial unitary air conditioning systems. Historically, this sector has used various ozone-depleting substances (ODS) such as CFCs and HCFCs as refrigerants. These ODS are being phased out under the Montreal Protocol and are being replaced with HFCs.

HFC emissions from the refrigeration and air conditioning sector result from the manufacturing process, from leakage over the operational life of the equipment, and from disposal at the end of the useful life of the equipment. These gases have 100-year global warming potentials (GWP), which are 140 to 11,700 times that of carbon dioxide, so their potential impact on climate change can be significant (Table 1). By the same token, any reductions of these gases can have a large potential benefit.

This protocol addresses emissions from manufacturing, operation, and disposal phases.

II. Choice of Activity Data and Emission Factors

HFC emissions can be estimated based on data readily available to manufacturers and users of refrigeration and air-conditioning equipment. This guideline contains a screening method to establish significance of HFC emissions and two quantification approaches to quantify HFC emissions. Reporters can choose a quantification approach based on data availability, purpose of quantification, and the level of accuracy required.

The recommended approach for equipment manufacturers and for equipment users who maintain their own equipment is to estimate HFC emissions based on the amount of refrigerant purchased and used. This “Sales-Based Approach” requires data that should be available from entity purchase and service records, and tracks emissions from equipment manufacturing (producers) or installation (users), operation, servicing, and disposal.

The recommended approach for equipment users who have contractors service their equipment is to track emissions at each stage of the lifecycle of the equipment. The “Lifecycle Stage Approach” requires information on the quantity of refrigerant used to fill new equipment during installation, the quantity of refrigerant used to service equipment, the quantity of refrigerant recovered from retiring equipment, and the full and proper charges of new and retiring equipment. If notified in advance of the need for this information, the contractor should be able to provide it. The approach tracks emissions from equipment manufacturing, operation, servicing, and disposal.

ded if a company needs to establish whether its emissions
ıt or not compared to its other emission sources. The screening
ault equipment emission factors or leak rates. This approach
of the two quantification methods provided in this tool as it is
aches. The leak rates can be highly uncertain. Using the
determines that its emissions from RAC equipment are
two quantification methods to estimate HFC emissions.

reenhouse Gases and Refrigerants Source
IPCC Second Assessment Report (1995)
IPCC Second Assessment Report (1995)
IPCC Second Assessment Report (1995)
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